PROJECT — What BAYMAX is (and isn't)

Framing lock (2026-08-24): BAYMAX is a **stepping-stone learning project**. It is not a research contribution, not a daily-driver personal assistant, and not the primary CV flagship. It's a shipped, evaluated, honestly-documented tool-use agent that demonstrates engineering competence and methodology discipline. That's it. That's the artifact.

Consolidated from prior `PROBLEM.md` + `WRITEUP_INFO.md` + `METHODOLOGY.md` (all removed 2026-08-24; content that mattered is here, rest was publication-review overhead).

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1. What BAYMAX is

A locally-runnable personal-task agent that:

1. Accepts text input describing a productivity task (calendar / notion / gmail / clipboard territory)
2. Selects and executes a sequence of tool calls against fake in-memory adapters (calendar, notion, gmail, clipboard)
3. Confirms, refuses, or requests clarification when needed
4. Logs every action with trace / task identifiers
5. Is evaluated against a scripted benchmark suite (v1: 50 scenarios; may grow modestly)

Backed by a fine-tuned Qwen 2.5 1.5B + LoRA served via MLX on a Mac (Marv-side) OR an OpenAI backend (baseline).

The **defining artifact** is the eval-driven agent + shipped repo, not any specific model win.

2. What BAYMAX is NOT

- **Not a daily-driver personal assistant** — fake adapters only, no OAuth to real Google / Notion, response quality too low to trust with real actions. Making it useful for real daily use would need 2-3 months more focused work (real adapters + bigger model + menu-bar UI + safety guardrails). Not on the roadmap.
- **Not a research contribution** — the "small model + SFT for tool use" question has been researched extensively (xLAM 2024, ToolLLaMA 2023, Gorilla 2023, NexusRaven 2023). BAYMAX is a competent execution of a well-explored area, not novel.
- **Not the primary CV flagship** — that's TADASHI-1 (robotics). BAYMAX is portfolio evidence of engineering + methodology discipline; TADASHI-1 is the research signal.
- **Not a workshop / paper submission target** — see §5 writeup plan.
- **Not benchmarked as "small model beats frontier"** — that framing was rejected. See §4 comparison framing.

3. What the project actually measures

Honest hypothesis (2026-08-24 framing):

> Given a small fine-tuned model (Qwen 1.5B + LoRA), trained on frontier-derived generic tool-use data (Salesforce/xLAM-function-calling-60k), and evaluated on BAYMAX-specific scenarios via an output-adaptation layer, characterize what fraction of a frontier baseline's task-success rate the fine-tune achieves and enumerate the failure modes.

Expected outcome: fine-tune underperforms gpt-4o-mini baseline. Current smoke-test result: 0.24 vs 0.52 baseline. Proper training runs may improve to 0.35-0.50 range. **This is the honest outcome — not a project failure.**

4. Methodology essentials

Stripped down from prior publication-review-flavored discipline. What still matters:

- **Author-split discipline**: Marv owns runtime / backend / adapter. Ronin owns data / training / eval. Neither influences the distributional shape of the other's work based on knowledge of their own. Prevents distributional overfitting; keeps the headline number honest.
- **External training data**: Salesforce/xLAM-function-calling-60k, version-pinned. NOT same-author-as-eval training scenarios.
- **Output adapter**: translates fine-tuned model emissions (xLAM format) to BAYMAX's `AgentResponse` shape. Designed from tool schema, not from eval scenarios.
- **Basic tiered eval**: in-distribution + small OOD subset. Report both; don't aggregate into a single number.
- **Basic statistical honesty**: sample size disclosed alongside every headline number. Comparison claims phrased proportionally to sample size.
- **Comparison framing**: use one of A/D from below; NEVER head-to-head "small beats frontier."

Defensible comparison framings

Framing	Comparison	Notes
A. Fine-tune vs its own base	Qwen 1.5B untuned vs Qwen 1.5B + fine-tune	Measures fine-tuning delta cleanly
C. Cost-normalized	gpt-4o-mini $/scenario vs local Qwen ≈ $0	"N% quality at 0% cost"
D. Transfer-learning characterization ⭐	Characterize how far external-data training gets on a domain-specific contract	Matches chosen approach; honest
~~E. Head-to-head~~	Fine-tune vs frontier	Rejected — specialist vs generalist unfair

Current headline claim: **A + D reported honestly**. Neither requires the fine-tune to win.

5. Writeup plan

No workshop paper chasing. Publication tier for BAYMAX:

- **arXiv preprint** — post regardless. Free, immediate, citation-able.
- **Blog post** — 2500-word technical writeup on personal site. Aim for /r/MachineLearning frontpage.
- **HuggingFace model card + adapter publication** — Ronin uploads the fine-tuned Qwen adapter with methodology writeup as the model card. Industry-visible artifact.
- **CV bullet** per Tadashi CV Segment 2 format: exact technical problem, exact stack, exact ownership, exact scale, exact result.

Everything above earns its keep for portfolio + Master's application evidence. NeurIPS/ICLR workshops or main-track submissions do not.

6. Honest limitations (for the writeup + repo)

When talking about BAYMAX publicly:

- Fake adapters only — no real integration with Google / Notion / Gmail / system clipboard
- Small model — Qwen 1.5B, tiny by 2026 standards
- Small eval — 50-200 scenarios; underpowered for strong statistical claims
- Single-user, single-machine scope
- No streaming, no multi-turn dialog, no state persistence across requests
- Expected fine-tune underperformance vs frontier baseline
- No production deployment story (uvicorn manual start; no menu bar app)

All of the above are documented deliberately, not hidden. See `DEFERRED.md` for the full list with revisit triggers.

7. Users

Two builders: Marv + Ronin. **Not real end-users.** Self-dogfooding for evaluation is optional; the eval scenarios are the primary quality signal.

8. Team ownership (brief)

Module	Owner
Agent core, inference service, telemetry, MLXBackend, output adapter	Marv
Eval harness, training pipeline, tool adapters, scenario authorship	Ronin
Shared: repo scaffolding, ARCHITECTURE.md, writeup, demo	both

Full ownership rules and PR protocol: OWNERSHIP.md.

9. Related docs

Doc	Purpose
ROADMAP.md	Realistic timeline to stepping-stone ship (~5 weeks from 2026-08-24)
ARCHITECTURE.md	System design
OWNERSHIP.md	Module ownership + PR rules
BACKEND_FLOW.md / .html	Backend + agent design diagrams (post-ADR 0011 target)
BENCHMARKING.md	Eval methodology + results (Ronin)
SCORING.md	Deterministic scoring rules (Ronin)
DEFERRED.md	Known limitations + not-implemented items
decisions/	ADR archive

10. Meta

This project is Marv's genuine engineering + methodology-discipline artifact for Canadian Master's applications (Dec 2026 - Feb 2027 deadlines) and industry interviews. It ships in stepping-stone scope by end of September 2026 and then Marv moves to the next `future_work/` project. Not designed to be evolved further post-ship.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.